LLM-Distilled Semantic Knowledge for Task-Aware Object Selection

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***Abstract*—The conventional object detection frameworks identify objects in a scene but are incapable of selecting a particular object for a specific task. This paper presents a training-free framework for selecting an object based on a given input query using an offline 47 KB distilled knowledge bank. The objects are detected using a YOLOv8-nano detector, and their scores are compared against the given task using cosine similarity between Sentence-BERT encoded embeddings. The fusion of semantic relevance scores and detection confidence produces a final ranked selection with a threshold below which no objects are selected. This approach is evaluated across all 14 COCO-Tasks. The proposed method operates entirely on a CPU.**

***Index Terms*—Task-aware object selection, YOLOv8-nano, Sentence-BERT, LLM distillation, semantic knowledge, COCO-Tasks, edge AI.**

I. Introduction

Choosing an object based on a given task is an important challenge in computer vision, as simply identifying an object does not reveal whether it is suitable for a specific task. Sawatzky *et al.* used a Gated Graph Neural Network (GGNN) for task-driven object detection; however, their approach requires task-annotated training data and can be computationally expensive [1].

We propose a training-free approach that uses LLM-generated semantic descriptions to build a compact knowledge bank. YOLOv8-nano detects objects on a CPU, while Sentence-BERT embeddings and cosine similarity are used to measure their suitability for a given task. A weighted score ranks the detected objects, with a threshold allowing the system to determine when no suitable object is present in the scene.

II. Literature Survey

A. Task-Driven Object Detection

Sawatzky *et al.* introduced the COCO-Tasks dataset containing approximately 40,000 images covering 14 manipulation and navigation tasks for task-appropriate object selection [1]. Their approach used a Gated Graph Neural Network (GGNN) to model detected objects as graph nodes and propagate scene context through iterative message passing. The method achieved a mean Average Precision (mAP) of 32.6%, outperforming classification baselines by 3.8 percentage points. However, detector class errors can introduce inference noise.

B. Visual Question Answering and Scene Understanding

Early Visual Question Answering (VQA) methods combined CNN-based visual features with language representations to answer natural-language questions about images [2], [3]. Bottom-up and top-down attention mechanisms further improved scene understanding by combining object-level visual features with linguistic context. Unlike VQA, task-driven object selection must produce a localized bounding box corresponding to the queried task.

C. Object Affordance Detection

CNN-based methods have been used to detect affordance regions, while GGNN-based approaches model how the suitability of an object for an action depends on surrounding objects and scene context [4]–[7]. Other approaches investigate the functional understanding of tools using 3D object models and human demonstrations. However, these methods are generally category-specific and have limited ability to generalize to arbitrary task queries during inference.

D. CLIP: Language-Vision Alignment

CLIP learns aligned visual and linguistic representations from large-scale image-text pairs and enables zero-shot visual classification using image-text similarity [8]. Task-aware extensions such as TaskCLIP apply similar alignment for task-based scenarios but typically require GPU inference. Their relatively high computational and power requirements can make them unsuitable for resource-constrained mobile robotic platforms.

TABLE I
COMPARISON OF THE PROPOSED APPROACH WITH EXISTING WORK

Aspect	Base Paper	Proposed Approach
Core method	Gated Graph Neural Network (GGNN)	LLM-distilled semantic cosine similarity
Feature type	Visual features using ResNet101 CNN	Textual semantic embeddings using Sentence-BERT
Scene context	Graph-based object relationship modelling	Threshold-based relevance filtering
Training required	Yes – supervised using COCO-Tasks annotations	No – zero-shot and training-free
Inference dependency	Needs all objects in the scene simultaneously	Each object is scored independently

E. Sentence-BERT for Semantic Matching

Sentence-BERT generates efficient sentence-level semantic embeddings using siamese network training on Natural Language Inference (NLI) and Semantic Textual Similarity (STS) datasets [9]. It produces 384-dimensional dense vectors that effectively capture semantic relationships between task descriptions and object-related text. This makes it suitable for lightweight semantic matching in edge-deployed task-aware object selection systems.

F. Knowledge Distillation for Edge Deployment

Knowledge distillation traditionally transfers knowledge from large teacher models to compact student models for efficient deployment [10], [11]. An offline distillation approach can instead query large language models and encode their structured semantic knowledge into compact representations. The proposed work applies this paradigm to task-aware object selection by encoding knowledge for 14 tasks and 80 COCO categories into only 47 KB of quantized embeddings.

G. YOLOv8-nano for Lightweight Object Detection

YOLOv8-nano provides real-time object detection across the 80 COCO categories with low computational and memory requirements [12]. Its CPU-only inference capability makes it suitable for edge devices and robotic platforms without dedicated GPUs. Therefore, it can serve as the lightweight object detection component in an edge-deployed task-aware object selection pipeline.

III. COMPARISON WITH EXISTING ART

The proposed approach differs from the base work by Sawatzky *et al.* in terms of its core methodology, feature representation, training requirement, and inference strategy, as summarized in Table I.

IV. METHODOLOGY

The proposed framework consists of four major stages: object detection, semantic scoring, weighted fusion, and final selection, as shown in Fig. 1.

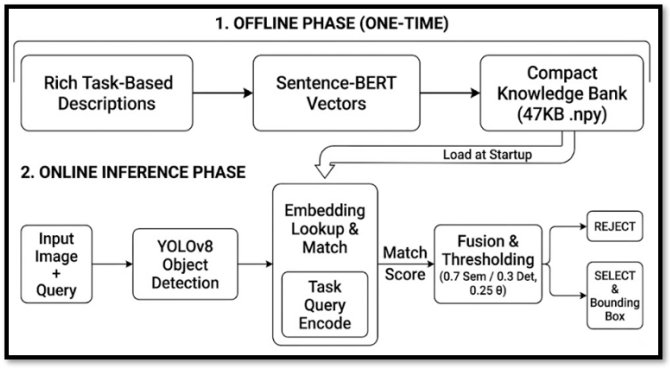


Fig. 1. Overall architecture of the proposed task-aware object selection framework.

A. Stage 1: Object Detection

The input image is processed using YOLOv8-nano. Objects having a detection confidence above a minimum threshold of 0.25 are retained along with their corresponding bounding-box coordinates.

B. Stage 2: Semantic Scoring

For each detected object, the corresponding precomputed semantic vector is retrieved from the knowledge bank. The input task query is encoded in real time using Sentence-BERT. Cosine similarity is then computed between the object embedding and the task embedding.

The semantic similarity score is expressed as

$$S_{\text{sem},i} = \frac{\mathbf{e}_{o_i} \cdot \mathbf{e}_t}{\|\mathbf{e}_{o_i}\| \, \|\mathbf{e}_t\|} \tag{1}$$

where $\mathbf{e}_{o_i}$ is the embedding of the ith detected object retrieved from the knowledge bank, and $\mathbf{e}_t$ is the real-time embedding of the task query string.

C. Stage 3: Weighted Fusion

The final score for each object incorporates both semantic relevance and detection reliability. The combined score can be represented as

$$S_i = \alpha S_{\text{sem},i} + (1 - \alpha) C_i \tag{2}$$

where $S_{\text{sem},i}$ represents the semantic relevance score, C_i represents the object detection confidence, and α is the weighting factor. Objects with a final score below 0.25 are excluded from consideration.

D. Stage 4: Selection and Output

The object having the highest final score above the specified threshold is selected as the output. Its corresponding bounding box is highlighted in the output image. If no detected object exceeds the threshold, the system determines that no suitable object is present for the given task.

V. Output

The proposed framework receives an input image and a task query. The output generated by the proposed framework is shown in Fig. 2.

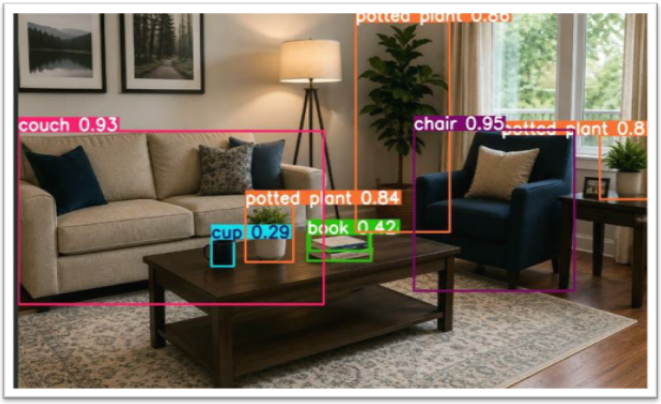


Fig. 2. Example output of the proposed task-aware object selection framework.

YOLOv8-nano first identifies the objects present in the image. The corresponding semantic embeddings are retrieved from the offline distilled knowledge bank and compared with the encoded task query using cosine similarity. The semantic relevance score and detection confidence are fused to rank the detected objects.

The highest-ranked object exceeding the threshold is selected and displayed with its corresponding bounding box. The threshold mechanism also enables the system to identify situations in which no suitable object is available for the specified task.

VI. Conclusion

This paper presents a lightweight and training-free framework for task-aware object selection. The proposed method combines YOLOv8-nano object detection with LLM-distilled semantic knowledge and Sentence-BERT-based cosine similarity to identify objects suitable for a given task. By fusing semantic relevance with detection confidence, the framework produces a ranked object selection while allowing no object to be selected when the relevance score is below a predefined threshold. The compact 47 KB knowledge bank and CPU-only operation make the proposed approach suitable for edge-deployed and resource-constrained robotic platforms.

Future work will focus on evaluating the framework across diverse real-world environments, optimizing the knowledge bank further, and extending the approach to additional object categories and task descriptions.

References

[1] Replace with the complete IEEE citation for Sawatzky et al., “Task-Driven Object Detection,” 2019.
[2] Replace with the complete IEEE citation for the first Visual Question Answering reference.
[3] Replace with the complete IEEE citation for the second Visual Question Answering or scene understanding reference.
[4] Replace with the complete IEEE citation for Object Affordance Detection reference 1.
[5] Replace with the complete IEEE citation for Object Affordance Detection reference 2.
[6] Replace with the complete IEEE citation for Object Affordance Detection reference 3.
[7] Replace with the complete IEEE citation for Object Affordance Detection reference 4.
[8] Replace with the complete IEEE citation for CLIP or TaskCLIP.
[9] Replace with the complete IEEE citation for Sentence-BERT.
[10] Replace with the complete IEEE citation for Knowledge Distillation reference 1.
[11] Replace with the complete IEEE citation for Knowledge Distillation reference 2.
[12] Replace with the complete IEEE citation for YOLOv8-nano.